WEEK 2 : LECTURE 2 - REQUIREMENTS & USE CASES

OBJECTIVES

- Requirements and Use Cases.
- Requirements capture.
- Requirements modelling.
- Use-cases.
- UML use-case diagrams

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REQUIREMENTS AND USE CASES.

- The aim of developing a new information system must be to produce something that meets the needs of the people who will be using it.
  - ✓ In order to do this, we must have a clear understanding both of the overall objectives of the business and of what it is that the individual users of the system are trying to achieve in their jobs.
- ✓ Unless you are in the rare position of developing a system for a new organization, you will need to understand how the business is operating at present and how people are working now.

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REQUIREMENTS AND USE CASES.

- Many aspects of the current system will need to be carried forward into the new system, so it is important that information about what people are doing is gathered and documented.
  - ✓ These are the requirements that are derived from the 'current system'.

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REQUIREMENTS AND USE CASES.

- The motivation for the development of new software is usually problems or inadequacies of the current system, so it is also essential to capture what it is that the users require of the new system that they cannot do with the existing system.
  - ✓ These are the 'new requirements'.
- The requirements should identify benefits of buying or building the new system that can be included in the 'Business Case' or 'Cost-Benefit Analysis', which is used to justify the expense of the new system in terms of the benefits that it will bring.

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REQUIREMENTS AND USE CASES

- 1. Current System
  - The existing system may be a:
    - ✓ manual one
    - ✓ based on paper documents
    - ✓ forms and files
    - ✓ an already computerized system
    - ✓ a combination of both manual and computerized elements

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REQUIREMENTS AND USE CASES

- 1. Current System
  - it is reasonably certain that large parts of the existing system meet the needs of the people who use it, that it has to some extent evolved over time to meet business needs and that users are familiar and comfortable with it.
  - It is almost equally certain that there are sections of the system that no longer meet the needs of the business, and that there are aspects of the business that are not dealt with in the existing system

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REQUIREMENTS AND USE CASES

- 1. Current System
  - It is important that the analyst gains a clear understanding of how the existing system works: parts of the existing system will be carried forward into the new one.

  - It is also important because the existing system will have shortcomings and defects, which must be avoided or overcome in the new system.
    - ✓ It is not always easy or possible to replace existing systems i.e. legacy systems

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REQUIREMENTS AND USE CASES

- 1. Current System
  - One approach to dealing with legacy systems is to create new front-ends, typically using modern graphical user interfaces and object-oriented languages, and wrap the legacy systems up in new software.
    - ✓ If this is the case, then it is also necessary to understand the interfaces to the legacy systems that the new wrappers will have to communicate with.

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REQUIREMENTS AND USE CASES

- 1. Current System
  - It is not always possible to leave legacy systems as they are and simply wrap them in new code.
  - Sometimes the suppliers of legacy systems withdraw support for older versions of applications; sometimes the technology in which they were written becomes obsolete and unsupported.
    - ✓ In these cases, the system may need to be replaced for IT reasons rather than because of business requirements for new functionality

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REQUIREMENTS AND USE CASES

- 1. Current System
  - Not everyone agrees that a detailed understanding of the current system is necessary.

    ✓ Advocates of Agile approaches to software development argue that the focus should be on understanding the users' requirements for the new system and not on the functionality of the old system.

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REQUIREMENTS AND USE CASES

- 1. Current System
  - These are reasons why investigating the existing system might be useful:
    - ✓ Some of the functionality of the existing system will be required in the new system.
    - ✓ Some of the data in the existing system is of value and must be migrated into the new system.
    - ✓ Technical documentation of existing computer systems may provide details of processing algorithms that will be needed in the new system.

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REQUIREMENTS AND USE CASES

- 1. Current System
  - These are reasons why investigating the existing system might be useful:
    - ✓ The existing system may have defects that we should avoid in the new system.
    - ✓ Studying the existing system will help us to understand the organization in general.
    - ✓ Parts of the existing system may be retained.

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REQUIREMENTS AND USE CASES

- 1. Current System
  - These are reasons why investigating the existing system might be useful:
    - ✓ We are seeking to understand how people do their jobs at present in order to characterize people who will be users of the new system.
    - ✓ We may need to gather baseline information against which we can set and measure performance targets for the new system.

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REQUIREMENTS AND USE CASES

- 2. New requirements
  - ✓ Whether you are investigating the working of the existing system or the requirements for the new system, the information you gather will fall into one of three categories:
    - ➢ ‘functional requirements’,
    - ➢ ‘non-functional requirements’
    - ➢ ‘usability requirements’.

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REQUIREMENTS AND USE CASES

- 2. New requirements
  - ✓ Functional requirements describe what a system does or is expected to do, often referred to as its functionality
  - ✓ This include:
    - ➢ descriptions of the processing that the system will be required to carry out;
    - ➢ details of the inputs into the system from paper forms and documents, from interactions between people, such as telephone calls, and from other systems;

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REQUIREMENTS AND USE CASES

- 2. New requirements
  - ✓ This include:
    - ➢ details of the outputs that are expected from the system in the form of printed documents and reports, screen displays and transfers to other systems;
    - ➢ details of data that must be held in the system.

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REQUIREMENTS AND USE CASES

- 2. New requirements
  - ✓ Non-functional requirements are those that describe aspects of the system that are concerned with how well it provides the functional requirements.
  - ✓ These include:
    - ➢ performance criteria such as desired response times for updating data in the system or retrieving data from the system;
    - ➢ ability of the system to cope with a high level of simultaneous access by many users;

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REQUIREMENTS AND USE CASES

- 2. New requirements
  - ✓ These include:
    - ➢ availability of the system with the minimum of downtime;
    - ➢ time taken to recover from a system failure;
    - ➢ anticipated volumes of data, either in terms of transaction throughput or of what must be stored;
    - ➢ security considerations such as resistance to attacks and the ability to detect attacks.

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REQUIREMENTS AND USE CASES

- 2. New requirements
  - ✓ Usability requirements are those that will enable us to ensure that there is a good match between the system that is developed and both the users of that system and the tasks that they will undertake when using it.
    - ➢ 'the extent to which specified users can achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use'.
  - ✓ Usability can be specified in terms of measurable objectives

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REQUIREMENTS AND USE CASES

- 2. New requirements
  - ✓ In order to build usability into the system from the outset, we need to gather the following types of information:
    - ➢ characteristics of the users who will use the system;
    - ➢ the tasks that the users undertake, including the goals that they are trying to achieve;
    - ➢ situational factors that describe the situations that could arise during system use;
    - ➢ acceptance criteria by which the user will judge the delivered system.

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REQUIREMENTS AND USE CASES

- There are five main fact-finding techniques that are used by analysts to investigate requirements
  - i. Background Reading
    - ✓ trying to gain an understanding of the organization.
    - ✓ These are the document to get information from:
      - ➢ company reports
      - ➢ organization charts
      - ➢ policy manuals
      - ➢ job descriptions
      - ➢ reports
      - ➢ documentation of existing systems.

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REQUIREMENTS AND USE CASES

- 1. Background Reading
  - ✓ Advantages of Background Reading
    - ➢ helps the analyst to get an understanding of the organization before meeting employees
    - ➢ allows the analyst to prepare for other types of fact finding;
    - ➢ existing system documentation may provide formally defined requirements for the current system.

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REQUIREMENTS AND USE CASES

- i. Background Reading
  - ✓ Disadvantage of Background Reading
    - ➢ Written documents often do not match up to reality;
      - o they may be out of date or they may reflect the official policy on matters that are dealt with differently in practice.

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REQUIREMENTS AND USE CASES

- ii. Interviews
  - ✓ most widely used fact-finding technique; it is also the one that
  - ✓ requires the most skill and sensitivity
  - ✓ Advantages of interviews
    - ➢ Personal contact allows the analyst to be responsive and adapt to what the user says. Because of this, interviews produce high-quality information

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REQUIREMENTS AND USE CASES

- ii. Interviews
  - ✓ Advantages of interviews
    - ➢ The analyst can probe in greater depth about the person's work than can be achieved with other methods.
    - ➢ If the interviewee has nothing to say, the interview can be terminated

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REQUIREMENTS AND USE CASES

- ii. Interviews
  - ✓ Disadvantages of interviews
    - ➢ Interviews are time-consuming and can be the most costly form of fact gathering.
    - ➢ Interview results require the analyst to work on them after the interview: the transcription of tape recordings or writing up of notes.
    - ➢ Interviews can be subject to bias if the interviewer has a closed mind about the problem.
    - ➢ If different interviewees provide conflicting information, it can be difficult to resolve later.

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REQUIREMENTS AND USE CASES

- ii. Interviews

Box 6.1 Guidelines on Interviewing

Conducting an interview requires good planning, good interpersonal skills and an alert and responsive frame of mind. These guidelines cover the points you should bear in mind when planning and conducting an interview.

Before the interview

You should always make appointments for interviews in advance. You should give the interviewee information about the likely duration of the interview and the subject of the interview.

Being interviewed takes people away from their normal work. Make sure that they feel that it is time well spent.

It is conventional to obtain permission from an interviewee's line manager before interviewing them. Often the analyst interviews the manager first and uses the opportunity to get this permission.

In large projects, an interview schedule should be drawn up showing who is to be interviewed, how often and for how long. Initially this will be in terms of the job roles of interviewees rather than named individuals. It may be the manager who decides which individual you interview in a particular role.

Have a clear set of objectives for the interview. Plan your questions and write them down. Some people write the questions with space between them for the replies.

Make sure your questions are relevant to the interviewee and his or her job.

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REQUIREMENTS AND USE CASES

- ii. Interviews

At the start of the interview

Introduce yourself and the purpose of the interview.

Arrive on time for interviews and stick to the planned timetable—do not overrun.

Ask the interviewee if he or she minds you taking notes or tape-recording the interview. Even if you tape-record an interview, you are advised to take notes. Machines can fail! Your notes also allow you to refer back to what has been said during the course of the interview and follow up points of interest.

Remember that people can be suspicious of outside consultants who come in with clipboards and stopwatches. The cost–benefit analyses of many information systems justify the investment in terms of savings in jobs!

During the interview

Take responsibility for the agenda. You should control the direction of the interview. This should be done in a sensitive way. If the interviewee is getting away from the subject, bring them back to the point. If what they are telling you is important, then say that you will come back to it later and make a note to remind yourself to do so.

Use different kinds of question to get different types of information. Questions can be open-ended—'Can you explain how you complete a timesheet?'—or closed—'How many staff use this system?'. Do not, however, ask very open-ended questions such as 'Could you tell me what you do?'.

Listen to what the interviewee says and encourage him or her to expand on key points.

Keep the focus positive if possible. Make sure you have understood answers by summarizing them back to the interviewee. Avoid allowing the interview to degenerate into a session in which the interviewee complains about everyone and everything.

You may be aware of possible problems in the existing system, but you should avoid prejudging issues by asking questions that focus too much on problems. Gather facts.

Be sensitive about how you use information from other interviews that you or your colleagues have already conducted, particularly if comments were negative or critical.

Use the opportunity to collect examples of documents that people use in their work, ask if they mind you having samples of blank forms and photocopies of completed paperwork.

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REQUIREMENTS AND USE CASES

- ii. Interviews

After the interview
Thank the interviewee for their time. Make an appointment for a further interview if it is necessary. Offer to provide them with a copy of your notes of the interview for them to check that you have accurately recorded what they told you.
Transcribe your tape or write up your notes as soon as possible after the interview while the content is still fresh in your mind.
If you said that you would provide a copy of your notes for checking then send it to the interviewee as soon as possible. Update your notes to reflect their comments.

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REQUIREMENTS AND USE CASES

- iii. Observations
  - Watching people carrying out their work in a natural setting can provide the analyst with a better understanding of the job than interviews, in which the interviewee will often concentrate on the normal aspects of the job and forget the exceptional situations and interruptions that occur & which the system will need to cope with.

  - Observation also allows the analyst to see what information people use to carry out their jobs.

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REQUIREMENTS AND USE CASES

- iii. Observations
  - This can tell you about the documents they refer to, whether they have to get up from their desks to get information, how well the existing system handles their needs.
  - Advantages of Observations
    - ✓ provides 1st-hand experience of the way that the current system operates.
    - ✓ Data is collected in real time & can have a high level of validity if care is taken in how the technique is used.

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REQUIREMENTS AND USE CASES

- iii. Observations
  - Advantages of Observations
    - ✓ used to verify information from other sources or to look for exceptions to the standard procedure.
    - ✓ Baseline data about the performance of the existing system and of users can be collected.
  - Disadvantages of Observations
    - ✓ Most people do not like being observed & are likely to behave differently from the way in which they would normally behave. This can distort findings and affect the validity.

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REQUIREMENTS AND USE CASES

- iii. Observations
  - Disadvantages of Observations
    - ✓ requires a trained and skilled observer for it to be most effective.
    - ✓ There may be logistical problems for the analyst; e.g., if the staff to be observed work shifts or travel long distances in order to do their job.
    - ✓ There may also be ethical problems if the person being observed deals with sensitive private or personal data or directly with members of the public; for example in a doctor's surgery.

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REQUIREMENTS AND USE CASES

- iv. Document Sampling
- Document sampling can be used in two different ways:
  - ✓ First, the analyst will collect copies of blank & completed documents during the course of interviews and observation sessions.
  - ✓ These will be used to determine the information that is used by people in their work and the inputs to and outputs from processes which they carry out, either manually or using an existing computer system.

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REQUIREMENTS AND USE CASES

- iv. Document Sampling
- Document sampling can be used in two different ways:
  - ✓ Second, the analyst may carry out a statistical analysis of documents in order to find out about patterns of data.
    - ➢ E.g many documents such as order forms contain a header section & a number of lines of detail

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REQUIREMENTS AND USE CASES

- iv. Document Sampling
  - ✓ The analyst may want to know the distribution of the number of lines in an order.
    - ➢ This will help later in estimating volumes of data to be held in the system and in deciding how many lines should be displayed on screen at one time

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REQUIREMENTS AND USE CASES

- iv. Document Sampling
  - Advantages of Document Sampling
    - ✓ Can be used to gather quantitative data, such as the average number of lines on an invoice and the range of values.
    - ✓ Can be used to find out about error rates in paper documents
  - Disadvantages of Document Sampling
    - ✓ If the system is going to change dramatically, existing documents may not reflect how it will be in future.

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REQUIREMENTS AND USE CASES

- v. Questionnaires
  - ✓ Questionnaires are a research instrument that can be applied to fact finding in system development projects.
  - ✓ They consist of a series of written questions.
  - ✓ The questionnaire designer usually limits the range of replies that respondents can make by giving them a choice of options.
  - ✓ YES/NO questions only give the respondent two options. (Sometimes a DON'T KNOW option is needed as well.)

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REQUIREMENTS AND USE CASES

- v. Questionnaires
  - ✓ YES/NO questions only give the respondent two options. (Sometimes a DON'T KNOW option is needed as well.)
  - ✓ Some questions do not have a fixed number of responses, and must be left open-ended for the respondent to enter what they like.
  - ✓ If you plan to use questionnaires for requirements gathering, they need very careful design.

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REQUIREMENTS AND USE CASES

- v. Questionnaires
- Advantages of Questionnaires
  - ✓ An economical way of gathering data from a large number of people.
  - ✓ If the questionnaire is well-designed, the results can be analysed easily, possibly by computer.
- Disadvantages of Questionnaires
  - ✓ Good questionnaires are difficult to construct.
  - ✓ Respondents may not honestly answer but just rush through the questionnaire

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REQUIREMENTS AND USE CASES

- v. Questionnaires

✓ Types of questions in a questionnaire

YES/NO Questions

Do you print reports from the existing system? YES NO 10
(Please circle the appropriate answer.)

Multiple Choice Questions

How many new clients do you obtain in a year? 11
(Please tick one box only.)

a) 1–10 ☐
b) 11–20 ☐
c) 21–30 ☐
d) 31+ ☐

Scaled Questions

How satisfied are you with the response time of the stock update?
(Please circle one option.)

1. Very satisfied 2. Satisfied 3. Dissatisfied 4. Very dissatisfied 12

Open-ended Questions

What additional reports would you require from the system?

____

____

____

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REQUIREMENTS AND USE CASES

- v. Questionnaires

✓ Questionnaires Guidelines

Box 6.2 **Guidelines on Questionnaires**

Using questionnaires requires good planning. If you send out 100 questionnaires and they do not work, it is difficult to get respondents to fill in a second version. These guidelines cover the points you should bear in mind when using questionnaires.

Coding

How will you code the results? If you plan to use an optical mark reader, then the response to every question must be capable of being coded as a mark in a box. If you expect the results to be keyed into a database for analysis, then you need to decide on the codes for each possible response. If the questions are open-ended, how will you collate and analyse different kinds of responses?

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REQUIREMENTS AND USE CASES

- v. Questionnaires

✓ Questionnaires Guidelines

Analysis

Whatever analysis you plan should be decided in advance. If you expect to carry out a statistical analysis of the responses, you should consult a statistician **before** you finalize the questions. Statistical techniques are difficult to apply to responses to poorly designed questions.

You can use a special statistical software package, a database or even a spreadsheet to analyse the data.

Piloting

You should try out your questionnaire on a small pilot group or sample of your respondents. This enables you to find out if there are questions they do not understand, they misinterpret or they cannot answer.

If you plan to analyse the data using statistical software, a database or a spreadsheet, you can create a set of trial data to test your analysis technique.

Sample size and structure

If you plan to use serious statistical techniques, then those techniques may place lower limits on your sample size. If you want to be sure of getting a representative sample, by age, gender, department, geographical location, job grade or experience of existing systems, then that will help to determine how many people to include. Otherwise it may be down to you to choose a sensible percentage of all the possible respondents.

Delivery

How will you get the questionnaires to your respondents, and how will they get their replies back to you?

You can post them, or use internal mail in a large organization, fax them, email them or create a web-based form on the intranet and notify your target group by email. If you use the intranet, you may want to give each respondent a special code, so that only they can complete their own questionnaire.

Your respondents can then post, fax or email their responses back to you, or complete them on the intranet.

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REQUIREMENTS AND USE CASES

- v. Questionnaires

✓ Questionnaires Guidelines

Respondent information

What information about the respondents do you want to gather at the same time as you collect their views and requirements? If you want to analyse responses by age, job type or location, then you need to include questions that ask for that information.

You can make questionnaires anonymous or you can ask respondents for their name. If the questionnaire is not anonymous, you need to think about confidentiality. People will be more honest in their replies if they can respond anonymously or in confidence.

If you ask for respondents' names and you store that information, then in the UK you should consider the provisions of the Data Protection Act (1998). (See also Chapter 12.) There are similar requirements in other countries.

Covering letter or email

In a covering letter you should explain the purpose and state that the questionnaire has management support. Give an estimate of the time required to fill in the questionnaire and a deadline for its return. Thank the respondents for taking part.

Structure

Structure the questionnaire carefully. Give it a title, and start with explanatory material and notes on how to complete it. Follow this with questions about the respondent (if required). Group questions together by subject. Avoid lots of instructions like: 'If you answered YES to Q. 7, now go to Q. 13'. Keep it reasonably short.

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REQUIREMENTS AND USE CASES

- v. Questionnaires

✓ Questionnaires Guidelines

Return rate
Not everyone will necessarily respond. You need to plan for this and either use a larger sample than you need or follow up with reminders. If you use a form on the intranet, you should be able to identify who has not responded and email them reminders. Equally, you can email a thank you to those who do respond.
Feedback
This needs to be handled carefully—telling everyone that 90% of the company cannot use the existing system may not go down well—but people do like to know what use was made of the response they made. They may have spent half an hour filling in your questionnaire, and they will expect to be informed of the outcome. A summary of the report can be sent out to branches, distributed to departments, sent to named respondents or placed on the intranet.

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REQUIREMENTS AND USE CASES

- vi. Other Techniques

✓ Some kinds of system require special fact-finding techniques.

✓ Expert systems are computer systems that are designed to embody the expertise of a human expert in solving problems.

✓ Examples include systems for medical diagnosis, stock market trading and geological analysis for mineral prospecting.

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REQUIREMENTS AND USE CASES

- vi. Other Techniques
- ✓ The process of capturing the knowledge of the expert is called knowledge acquisition and, as it differs from establishing the requirements for a conventional information system, a number of specific techniques are applied.
- ✓ Some of these are used in conjunction with computer-based tools.

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REQUIREMENTS AND USE CASES

- User involvement
  - ✓ The success of a systems development project depends not just on the skills of the team of analysts, designers and programmers who work on it, or on the project management skills of the project manager, but on the effective involvement of users in the project at various stages of the lifecycle
  - ✓ Stakeholders include all people who stand to gain (or lose) from the implementation of the new system: users, managers and budget-holders.

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REQUIREMENTS AND USE CASES

- User involvement
  - ✓ This will include the following categories of people:
    - ➢ senior management—with overall responsibility for running the organization
    - ➢ financial managers with budgetary control over the project
    - ➢ managers of the user department(s)
    - ➢ representatives of the IT department delivering the project
    - ➢ representatives of users.

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REQUIREMENTS AND USE CASES

- User involvement
  - ✓ Users will be involved in different roles during the course of the project as:
    - ➢ subjects of interviews to establish requirements
    - ➢ representatives on project committees
    - ➢ those involved in evaluating prototypes
    - ➢ those involved in testing
    - ➢ subjects of training courses
    - ➢ end-users of the new system.

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REQUIREMENTS AND USE CASES

- Documenting Requirements
  - ✓ Information systems professionals need to record facts about the organization they are studying and its requirements.
  - ✓ As soon as the analysts start gathering facts, they will need some means of documenting them.
  - ✓ In the past the emphasis was on paper forms, but now it is rare for a large scale project to depend on paper-based documentation

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REQUIREMENTS AND USE CASES

- Documenting Requirements
  - ✓ systems analysts and designers model the new system in a mixture of diagrams, data and text.
  - ✓ The important thing to bear in mind is that within a project some set of standards should be adhered to e.g. UML
  - ✓ But there will also be other kinds of documents, not all of which fit into the UML framework.
  - ✓ required to keep track of these documents and ensure that they are stored safely and in a way that enables them to be retrieved when required.

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REQUIREMENTS AND USE CASES

- Documenting Requirements
  - ✓ Such documents include:
    - ➢ records of interviews and observations
    - ➢ details of problems
    - ➢ copies of existing documents and where they are used
    - ➢ details of requirements
    - ➢ details of users
    - ➢ minutes of meetings.

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REQUIREMENTS AND USE CASES

- Use Cases
  - ✓ Use cases are descriptions of the functionality of the system from the users' perspective.
  - ✓ Use case diagrams are used to show the functionality that the system will provide and to show which users will communicate with the system in some way to use that functionality

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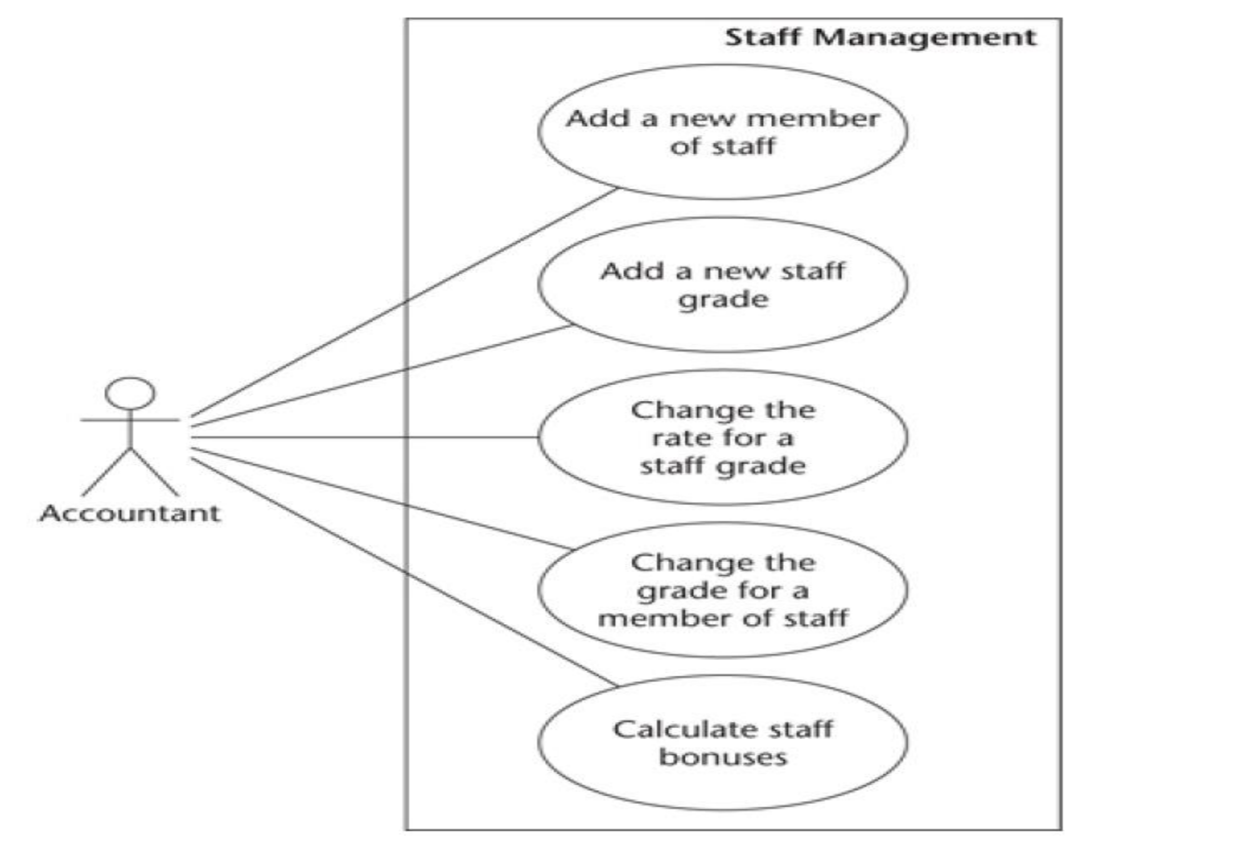
REQUIREMENTS AND USE CASES

- Use Cases



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REQUIREMENTS AND USE CASES

- Use Cases
  - ✓ Use cases are supported by behavior specifications.
  - ✓ These specify the behaviour of each use case either using UML diagrams, such as activity diagrams, communication diagrams or sequence diagrams, or in text form as use case descriptions
  - ✓ Textual use case descriptions provide a description of the interaction between the users of the system, termed actors, and the high-level functions within the system, the use cases

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REQUIREMENTS AND USE CASES

- Use Cases
  - ✓ These descriptions can be in summary form or in a more detailed form in which the interaction between actor and use case is described in a step-by-step way.
  - ✓ Whichever approach is used, it should be remembered that the use case describes the interaction as the user sees it, and is not a definition of the internal processes within the system or some kind of program specification

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REQUIREMENTS AND USE CASES

- Use Cases
  - ✓ Notation: Use case diagrams show three aspects of the system: actors, use cases and the system or subsystem boundary.
  - ✓ Actors represent the roles that people, other systems or devices take on when communicating with the particular use cases in the system
  - ✓ one actor can represent several people or job titles
  - ✓ Equally, a particular person or job title may be represented by more than one actor on use case diagrams

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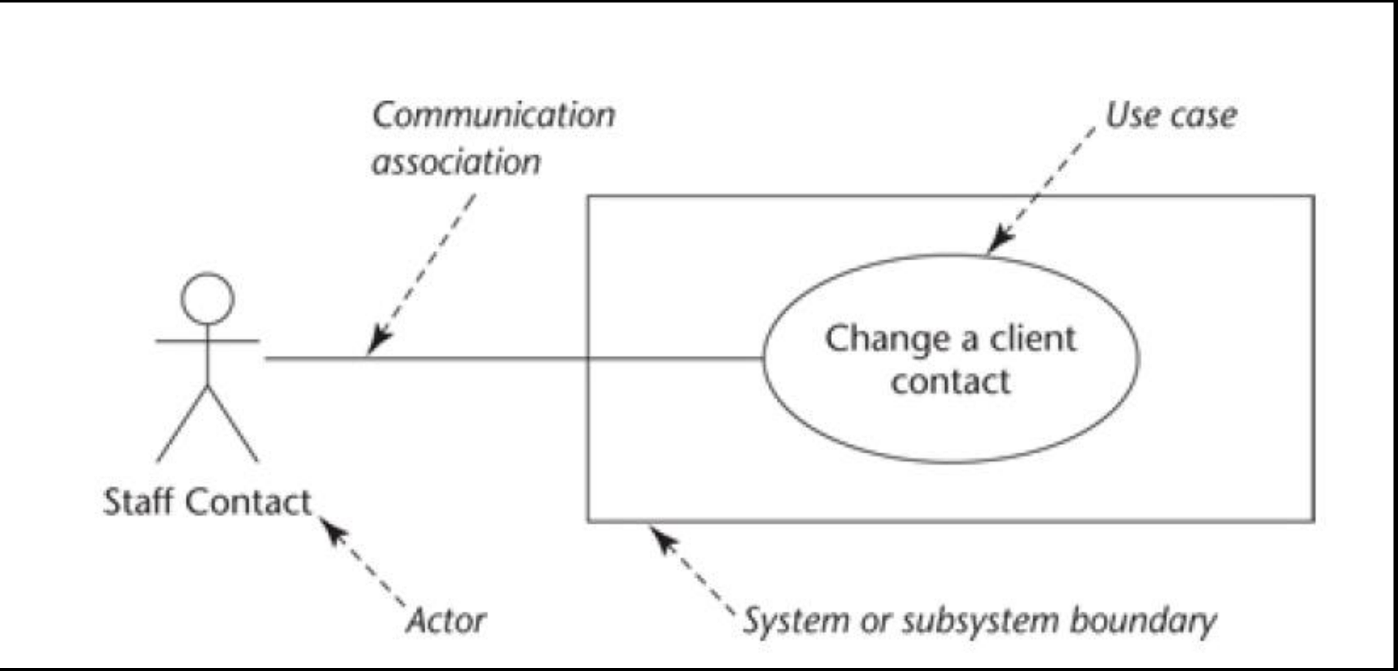
REQUIREMENTS AND USE CASES

- Use Cases
  - ✓ Use case notation



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REQUIREMENTS AND USE CASES

- Use Cases
  - ✓ Use case for Campaign Manager Actor

Figure 6.6 Use case showing Campaign Manager actor.

The use case description associated with each use case can be brief:

Assign staff to work on a campaign

The campaign manager selects a particular campaign. A list of staff not already working on that campaign is displayed, and he or she selects those to be assigned to this campaign.

Alternatively, it can provide a step-by-step breakdown of the interaction between the user and the system for the particular use case. An example of this extended approach is provided below.

Assign staff to work on a campaign

Actor Action	System Response
1. The actor enters the client name.	2. Lists all campaigns for that client.
3. Selects the relevant campaign.	4. Displays a list of all staff members not already allocated to this campaign.
5. Highlights the staff members to be assigned to this campaign.	6. Presents a message confirming that staff have been allocated.

Alternative Courses

Steps 1–3. The actor knows the campaign name and enters it directly.

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REQUIREMENTS AND USE CASES

- Use Cases
  - ✓ One way of documenting use cases is to use a template including the following sections:
    - ➢ name of use case
    - ➢ pre-conditions (things that must be true before the use case can take place)
    - ➢ post-conditions (things that must be true after the use case has taken place)
    - ➢ purpose (what the use case is intended to achieve)
    - ➢ description (in summary or in the format above).

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REQUIREMENTS AND USE CASES

- Use Cases
  - ✓ One of the benefits of developing use cases as part of the specification of the system that is to be implemented, is that they can form the basis of scenarios that can be used as test cases when the system has been developed.

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REQUIREMENTS CAPTURE AND MODELLING

- The first stage of most projects is one of capturing and modelling the requirements for the system
- Let us look at an example or case study

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REQUIREMENTS CAPTURE AND MODELLING

Case Study Example

You have already seen several examples from the case study in this chapter. The use cases are determined by the analyst from the documentation that is gathered from the fact-finding process. What follows is a short excerpt from an interview transcript, which has been annotated to show the points that the analyst would pick up on and use to draw the use case diagrams and produce the associated documentation. The interview is between Dave Harris, a systems analyst, and Peter Bywater, an Account Manager at Agate. It is from one of the interviews with the objective: 'To establish additional requirements for new system' (in the fact-finding plan in the earlier case study section in this chapter). Commentary has been added in brackets.

Dave Harris: *You were telling me about concept notes. What do you mean by this?*
Peter Bywater: *At present, when we come up with an idea for a campaign we use a word processor to create what we call a concept note. We keep all the note files in one directory for a particular campaign, but it's often difficult to go back and find a particular one.*
DH: So is this something you'd want in the new system?
PB: Yes. We need some means to enter a concept note and to find it again. *(This sounds like two possible use cases. Who are the actors?)*
DH: So who would you want to be able to do this?
PB: I guess that the staff working on a campaign should be able to create a new note in the system.
DH: Only them? *(Any other actors?)*
PB: Yes, only the staff actually working on a campaign.
DH: What about finding them again? Is this just to view them or could people modify them?
PB: Well, we don't change them now. We just add to them. It's important to see how a concept has developed. So we would only want to view them. But we need some easy way of browsing through them until we find the right one. *(Who are the actors for this?)*
DH: Can anyone read the concept notes?
PB: Yes, any of the staff might need to have a look.
DH: Would you need any other information apart from the text of the concept itself? *(Thinking ahead to Chapter 7!)*
PB: Yes. It would be good to be able to give each one a title. Could we use the titles then when we browse through them? Oh, and the date, time and whoever created that concept note.
DH: Right, so you'd want to select a campaign and then see all the titles of notes that are associated with that campaign, so you could select one to view it? *(Thinking about the interaction between the user and the system.)*
PB: Yes, that sounds about right .

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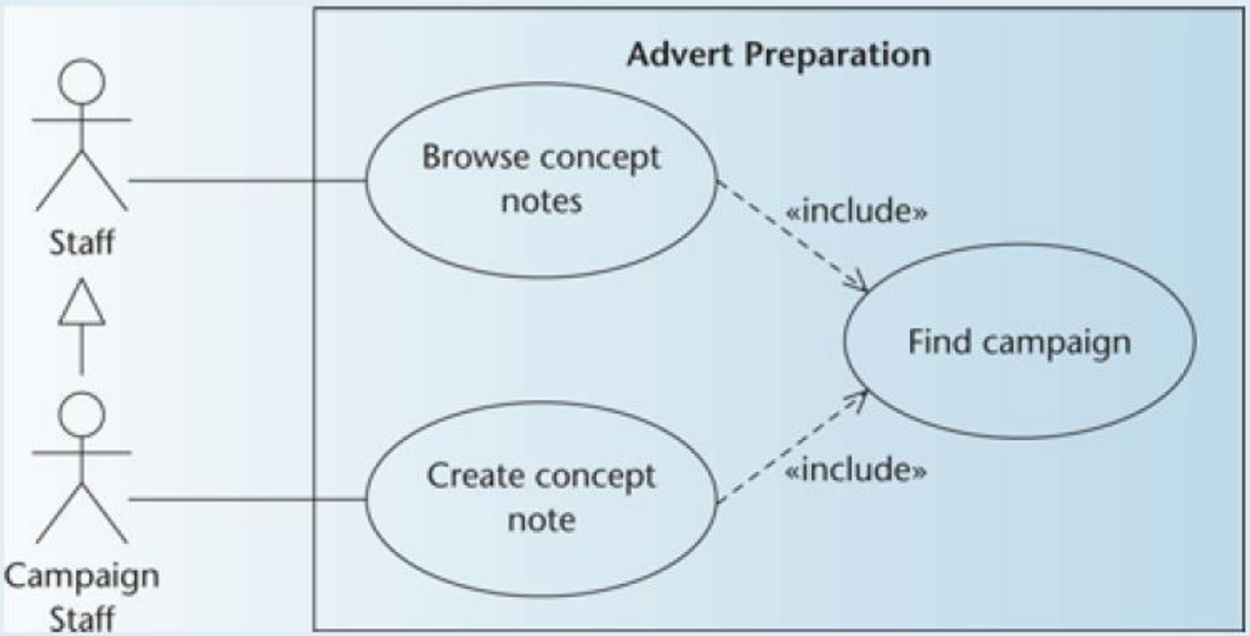
REQUIREMENTS CAPTURE AND MODELLING

(From this information, Dave Harris is going to be able to develop the use case descriptions for two use cases:

Create concept note

Browse concept notes

The use case diagram is shown in Fig. 6.16. The use case descriptions will be as follows.)



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REQUIREMENTS CAPTURE AND MODELLING

Create concept note
A member of staff working on a campaign can create a concept note, which records ideas, concepts and themes that will be used in an advertising campaign. The note is in text form. Each note has a title. The person who created the note, the date and time are also recorded.

Browse concept notes
Any member of staff may view concept notes for a campaign. The campaign must be selected first. The titles of all notes associated with that campaign will be displayed. The user will be able to select a note and view the text on screen. Having viewed one note, others can be selected and viewed.

(The interaction here is quite straightforward, so we shall not need a more detailed breakdown of the interaction between user and system.

Note that in Fig. 6.16, because Campaign Staff is a specialization of Staff, we do not need to show a communication association between the Campaign Staff actor and the Browse concept notes use case.)

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THE END

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